

E05 CSTR Energy Balance Example: Excel part

Decision Guide: How Do I Change Reactor Temperature?

When using the $G(T)$ - $R(T)$ method, there are two primary ways to modify reactor operation:

Option 1: Change the Feed Temperature (T_0)

Ask yourself: Do I want to shift the $R(T)$ line left or right?

If YES: Change T_0

Increase T_0

Increase Feed Temperature

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Shift $R(T)$ Right

↓

Higher Operating Temperature

Decrease T_0

Decrease Feed Temperature

↓

Shift $R(T)$ Left

↓

Lower Operating Temperature

Key Idea

Changing T_0

↓

Shifts $R(T)$ without significantly changing its slope.

Option 2: Change the Heat Exchanger (UA)

Ask yourself: Do I want to change how aggressively heat is removed?

If YES: Change UA

Increase UA

Better Heat Removal



Steeper $R(T)$



Cooler Reactor

Decrease UA

Less Heat Removal



Flatter $R(T)$



Hotter Reactor

Key Idea

Changing UA



Rotates $R(T)$ by changing its slope.

Quick Troubleshooting Guide

Reactor Temperature Too High?

Try: Lower T_0 or Increase UA

Reactor Temperature Too Low?

Try: Increase T_0 or Decrease UA

Want Better Temperature Control?

Increase UA. A larger UA allows the heat exchanger to remove heat more effectively.

Most Important Rule

Feed Temperature



Moves $R(T)$

Heat Exchanger (UA)



Rotates $R(T)$

When analyzing non-isothermal CSTRs, this simple rule provides a quick way to predict how operating conditions will change the reactor's steady-state temperature.